$n_b(K, t+1) = n(K,t)$ [n_b = number of "b" nodes with given in-degree]

$+ \, m \, \Pi_b(K-1,t) \, n_b(K-1,t)$ [promotion opportunities available · prob of promotion given "#citation" · "paper type" · "number of these papers"]

$- \, m \, \Pi_b(K,t) \, n_b(K,t)$

$+ \, m(1-g)\delta_{K,0}$ (1) [if $K=0$ & node type is "b" then the new node is also added to the count]

$\Pi_{b,i}$ = prob("b" node recieves new in-degree from the node generated) $= \dfrac{(1-g)hK^{(in)} + g(1-h)K^{(in)} + \mu_b}{Z_b(t)} = \dfrac{[(1-g)h + g(1-h)]K^{(in)} + \mu_b}{Z_b(t)} = \dfrac{\lambda_b K + \mu_b}{Z_b(t)}$ (2)

[let $Z_b(t)$ be b normalisation constant contextualised by eqn (3)]

[let $K^{(in)} = K$ for notation simplicity]

$\sum_{i\in U}^{N} (\Pi_{b,i} + \Pi_{a,i}) = 1$ (3) [new node will necessarily create a new in-degree]

for simplicity: let :

i) $N(t_{init}) =' t_{init}$

ii) $E^{tot,in}(t_{init}) = E_a^{in} + E_b^{in} = m\,t_{init} = E_a^{in} + E_b^{in}$

iii) $Z_a(t) = Z_b(t)$ since constraint is only from (3) [$\therefore$ to allow us to analytically solve this conjecture is a helpful ansatz ((*) I think we fall down here)]

$\Rightarrow N_b(t_{init}) = (1-g)\,N(t_{init})$ & $E_b^{in}(t_{init}) = m\lambda_b t_{init}$

[$\lambda_a + \lambda_b = 1$ can be shown easily]

$\therefore$ via linearity of algorithm (? *, not sure this is right either)

$N_b(t) = (1-g)t$ & $E_b(t) = \lambda_b m t$

$\sum_{i\in V_b}^{N_b} \Pi_{b,i} + \sum_{i\in V_a}^{N_a} \Pi_{a,i} = 1$

$\sum_{i\in V_b}^{N_b} \Pi_{b,i} = \frac{1}{Z(t)}[\lambda_b \sum_{i\in V_b}^{N_b} K + \mu_b \sum_{i\in V_b}^{N_b} (1)] = \frac{1}{Z(t)}[\lambda_b E_b^{in} + \mu_b N_b] = \frac{1}{Z(t)}[\lambda_b^2 m t + \mu_b(1-g)t]$

& by symmetry, conjecture : $\sum_{i\in V_a}^{N_a} \Pi_{a,i} = \frac{1}{Z(t)}[\lambda_a^2 m t + \mu_a g t]$

$\Rightarrow Z(t) = [(\lambda_a^2 + \lambda_b^2)m + (g\mu_a + (1-g)\mu_b)]t \Rightarrow$ let $Z = [(\lambda_a^2 + \lambda_b^2)m + (g\mu_a + (1-g)\mu_b)]$, $Z \in \mathbb{R}$

& its also useful now to motivate $\tilde{Z} = m/Z$, $\tilde{Z} \in \mathbb{R}$

$n_b(K,t) = n_b(K,t) + \tilde{Z}t^{-1}(\lambda_b(K-1) + \mu_b)n_b(K-1,t) - \tilde{Z}t^{-1}(\lambda_b(K-1) + \mu_b)n_b(K,t) + m(1-g)\delta_{K,0}$ (4)

$p_b(K,t)$ = prob ("b" node has $K^{(in)} = K$) $= n_b(K,t)/N_b(t)$

$(1-g)[(t+1)p_b(K,t+1) - t p_b(K,t)] = \tilde{Z}(1-g)[\lambda_b(\lambda_b(K-1) + \mu_b)p_b(K-1,t) - (\lambda_b K + \mu_b)p_b(K,t)] + m(1-g)\delta_{K,0}$ (5)